STAT 692 Penguin Identification Project

Presentation Script (Approximately 10 Minutes)

Slide 1: Penguin Identification Project

“Hello Dr. Dr. Sharma and team. I’m Spencer Andries your statistical consultant for this project, and today I’ll be going over the results of your requested Penguin identification.

To reiterate your request, you wanted to identify the species of an unlabeled penguin specimen using body measurements from the Palmer Penguins dataset.”

Slide 2: Consulting Problem

“Specifically, you discovered a preserved penguin specimen that had lost its identification tag.

You had explained that the specimen had two known measurements: a bill length of 45 millimeters and a flipper length of 220 millimeters.

And wanted three questions answered:

First and foremost, you wanted to know what species is the specimen most likely to be?

Second, how confident were we in that classification?

And lastly, how does the specimen compare visually to known penguin species?

I won’t dive to deeply into this but because the problem was identifying the penguin, this became a classification problem which dictated some later decision that you will see.”

Slide 3: Dataset Overview

“As previously stated, you requested we use Palmer Penguins dataset for the analysis.

This dataset contains data on three penguin species: Adelie, Chinstrap, and Gentoo.

The original dataset contained 344 observations and 8 variables, including measurements such as bill length, bill depth, flipper length, body mass, sex, and island location.

Slide 4: Data Cleaning

For this analysis, I focused on bill length and flipper length because they were available for the unknown specimen and are biologically meaningful measurements. Which is why we removed all other columns and were left with species, bill length and flipper length"

"Before beginning the analysis, I examined the dataset for missing values.

The original data contained a small number of missing observations in the bill length and flipper length variables.

Because these variables were essential to the classification task, I removed rows with missing values.

After cleaning, the final dataset contained 342 penguins and no missing values in the variables used for analysis.

This was to ensure that all distance calculations used by the classification model would be based on complete information."

Slide 5: Exploratory Data Analysis

"Also exploring the distribution of observations across species.

The cleaned dataset contained 151 Adelie penguins, 68 Chinstrap penguins, and 123 Gentoo penguins.

You will notice sample sizes are not perfectly balanced between species, but the imbalance is relatively small.

This could affect the analysis but given that the group sizes are relatively similar and there is clear separation between species, I decided the imbalance would not meaningfully affect the results."

Slide 6: Summary Statistics

“Looking at the summary statistics.

Adelie penguins had the shortest average bill lengths and flipper lengths.

Chinstrap penguins had the longest average bill lengths but only moderately large flippers.

Gentoo penguins stood out because of their much longer flippers, averaging approximately 217 millimeters.

The unknown specimen had a flipper length of 220 millimeters, which is very close to the Gentoo average and substantially larger than the averages for Adelie and Chinstrap penguins.

So Even before fitting a model, this suggested that Gentoo was a likely candidate.”

Slide 7: Visualization Findings

“One of the most informative parts of the analysis was the scatterplot of bill length versus flipper length.

Each point represented a known penguin, and the unknown specimen was displayed as a black star.

As you can see the species formed fairly distinct clusters.

The unknown specimen fell directly within the cluster of Gentoo penguins and was visually separated from the main Adelie and Chinstrap groups.

Which provided strong evidence that the specimen belonged to the Gentoo species”

Slide 8: Why KNN?

“At this point I needed to choose an appropriate statistical method.

As this was primarily an identification problem instead of comparing the groups I selected K-Nearest Neighbors, or KNN instead of something like ANOVA.

KNN is a classification method that predicts the class of an observation by examining the classes of nearby observations.

It is fairly intuitive and made sense for your question.

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Slide 9: KNN Workflow

“The KNN workflow consisted of four main steps.

First, I selected bill length and flipper length as the predictor variables.

Second, I standardized the variables using the scale function. This ensured that bill length and flipper length contributed equally to the distance calculations.

Third, I calculated distances between the unknown specimen and all known penguins in the dataset.

Finally, I classified the specimen using the species labels of its 15 nearest neighbors.

Given the sample size 15 was a sensible choice”

Slide 10: Assumptions and Limitations

“Like any statistical method, KNN relies on several assumptions.

The most important assumption is that observations with similar measurements tend to belong to the same species.

The scatterplot we discussed earlier suggested this assumption was reasonable because the species formed distinct clusters.

I also standardized the predictors to ensure flipper didn’t impact the classification more than bill length.

Finally, the assumption that the observations are independent, was satisfied because each row represented a different penguin.

Slide 11: Results

“The model classified the unknown specimen as a Gentoo penguin.

In addition, all 15 of the specimen’s nearest neighbors were also Gentoo penguins.

This produced a KNN confidence measure of 100 percent.

While this does not mean literally 100 percent certainty that the specimen is Gentoo.

Taken together with the visualization and summary statistics, the evidence strongly supports classifying the specimen as a Gentoo penguin.”

Slide 12: Conclusions

“In conclusion, we can be confident in our final classification of Gentoo for the specimen you provided based on the available data.

Evidence:

- Similar body measurements.
- Placement within the Gentoo cluster.
- Closest matches were all Gentoo specimens.

Recommendation:

- Label specimen as Gentoo.

If you have any final question let me know otherwise I would like to thank you for choosing me to be your statistical consultant on this project and would love to work with you on any statistical needs you have in the future.

Have a great day!”